ENTERPRISE CALENDAR INFRASTRUCTURE

Slotbook

A scheduling platform built for zero double bookings under concurrency



PostgreSQL GiST exclusion

Fastify 5 on Node 24

React 19

02 THE PROBLEM

Why check then insert fails

Naive check then insert

A reads: 0 of 1 seat used

B reads: 0 of 1 seat used

A inserts a booking

B inserts a booking

2 bookings for 1 seat

Slotbook

A locks the host row

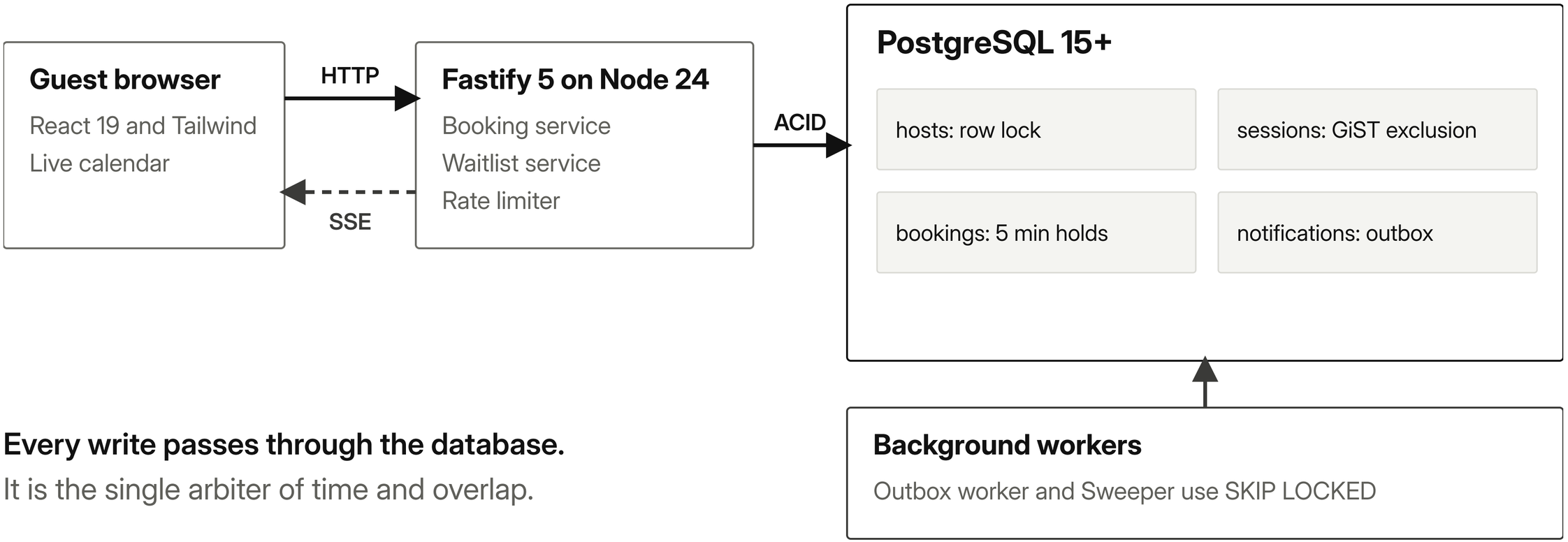
B waits for the lock

A inserts and commits

B gets 23P01 then HTTP 409

1 booking for 1 seat

Three tiers and one arbiter

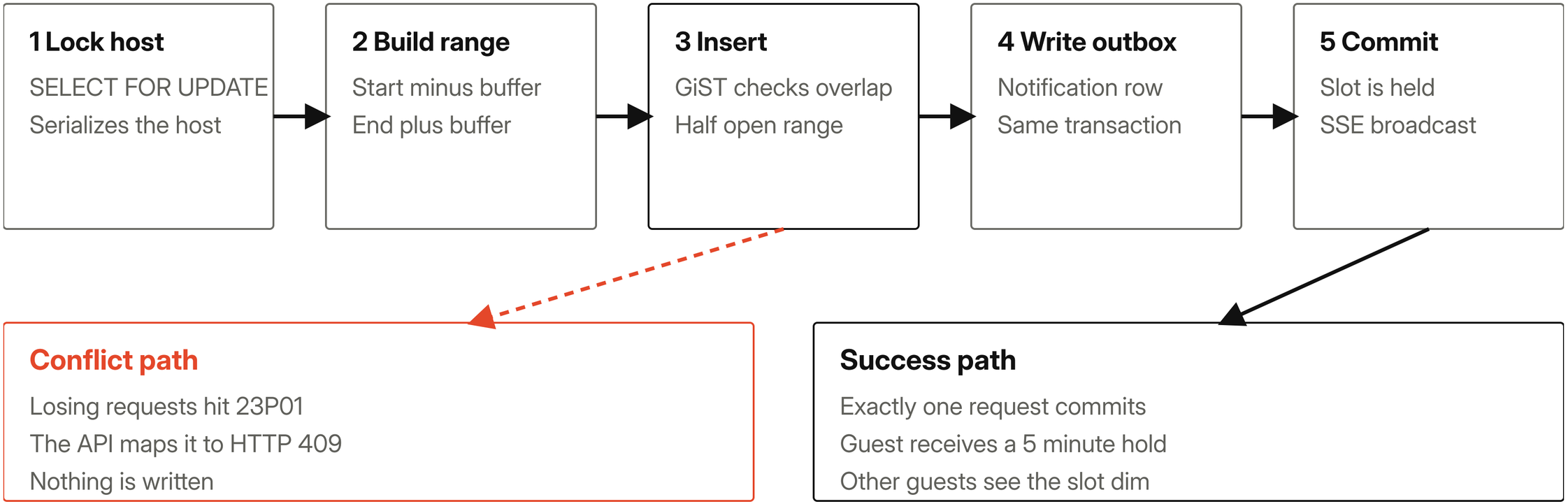


Every write passes through the database.

It is the single arbiter of time and overlap.

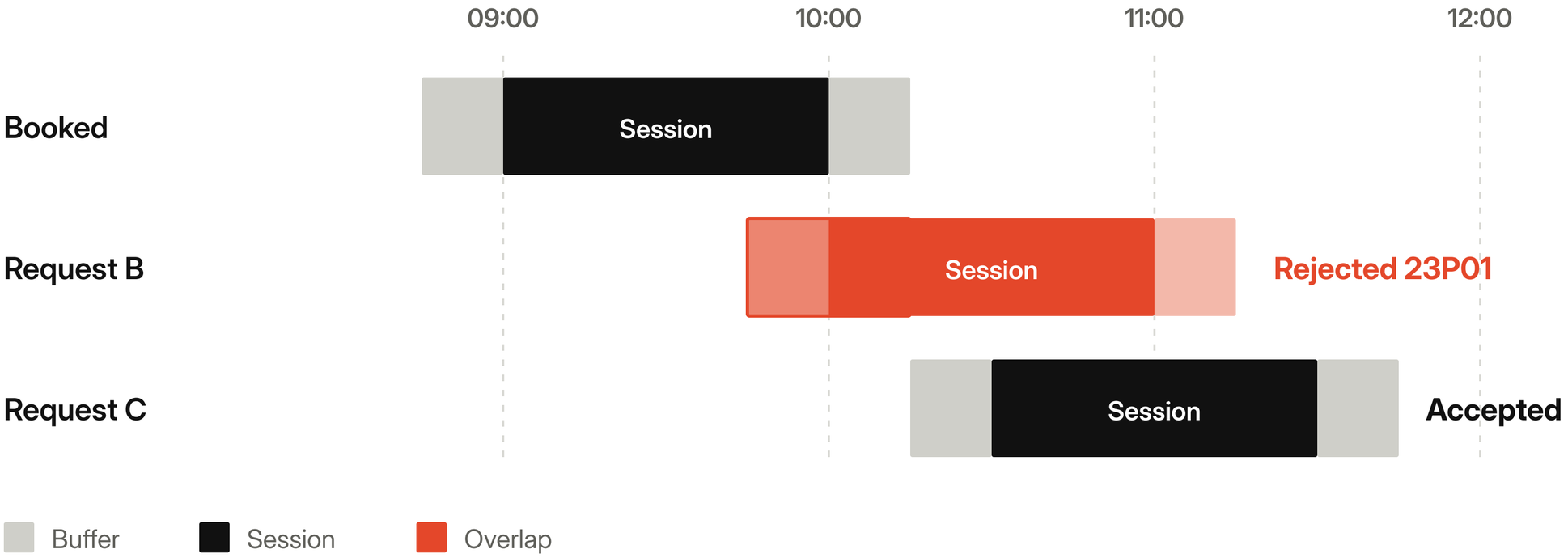
04 BOOKING FLOW

One request through the database



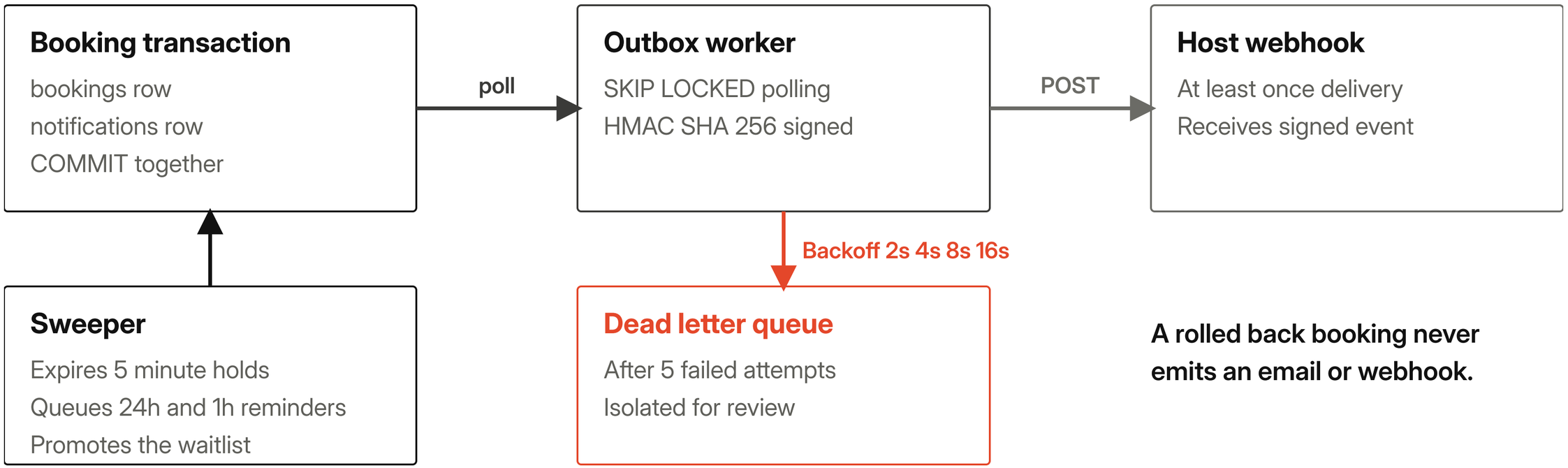
05 OVERLAP PREVENTION

Buffers are part of the reserved range



blocked_range = [start minus buffer before, end plus buffer after)

Transactional outbox and workers



A rolled back booking never emits an email or webhook.

07 VERIFIED RESULTS

Measured under race conditions

47 / 47
Automated tests passed

28 ms
Slot generation. P95 is 44 ms

0.00%
Error rate under race

1 of 50
Holds succeed. The other 49 receive HTTP 409

Zero double bookings is enforced by the database. Application code cannot bypass it.